How-To Library

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Troubleshooting Private Clusters

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Abstract

This article provides troubleshooting information for environments that use private clusters, including environments that use AWS PrivateLink or Azure Private Link.

Supported Versions

- Intelligent Data Management Cloud October 2026

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Troubleshooting Private Clusters

Review the following questions and answers to help you troubleshoot issues when you run IDMC on a private cluster.

The connection is timing out between the Secure Agent VM and the EKS cluster API.

The EKS cluster security group is blocking inbound traffic from the agent VM's network interface.

To resolve this issue, perform the following steps:

1. Identify the security group ID attached to your agent VM.
2. Navigate to the EKS cluster security group associated with the control plane.
3. Add the following inbound rule:

Property	Value
Type	HTTPS (Port 443)
Protocol	TCP
Source	Custom -> <Secure Agent VM security group ID>

I'm getting "Agent VM Authentication Failure (401 Unauthorized)." The Agent VM can see the cluster but can't perform operations like listing PODs or starting mappings.

The IAM role assigned to the agent VM isn't recognized by the EKS cluster's identity provider.

To resolve this issue, perform the following steps:

1. Create an EKS access entry specifically for the IAM role associated with your agent VM.
2. Associate the **AmazonEKSClusterAdminPolicy** or a custom restrictive policy with this access entry. This allows the Agent VM to interact with the Kubernetes API using its own IAM identity.

Operations like assuming roles are failing even though an STS VPC endpoint is present and the NAT gateway has been removed.

The AWS SDK/CLI on the agent VM is defaulting to the global STS Endpoint (sts.amazonaws.com). This fails in a private VPC because there's no internet access.

To resolve this issue, perform the following steps:

1. Force the agent VM to use the regional STS endpoint by setting the following environment variables on the VM:

```
export AWS_STS_REGIONAL_ENDPOINTS=regional
export AWS_DEFAULT_REGION=<your-region-id>
```

2. Verify with nslookup that `* nslookup sts.<region>.amazonaws.com` now returns the private IP of your VPC endpoint. STS calls no longer require a NAT gateway.

I'm receiving the following errors because my custom policy has different roles that don't have permission to start the required plugins in the EKS cluster:

```
Jan 30 11:45:48 ip-170-0-137-165.us-west-2.compute.internal kubelet[3541]: E0130
11:45:48.345462    5804 main.go:262] Error running credential provider plugin: operation
error ECR: GetAuthorizationToken, https response error StatusCode: 400, RequestID:
4e97f050-7da5-4c66-a3d4-cd115e223425, api error AccessDeniedException: User:
arn:aws:sts::543463116864:assumed-role/gani-eks-nodegroup-role/i-064444757ad7c9e50 is
not authorized to perform: ecr:GetAuthorizationToken on resource: * because no VPC
endpoint policy allows the ecr:GetAuthorizationToken action
Jan 30 11:45:48 ip-170-0-137-165.us-west-2.compute.internal kubelet[3541]: E0130
11:45:48.346506    3541 pod_workers.go:1301] "Error syncing pod, skipping" err="failed
to \"StartContainer\" for \"kube-proxy\" with ImagePullBackOff: \"Back-off pulling image
\\\"602401143452.dkr.ecr.us-west-2.amazonaws.com/eks/kube-proxy:v1.33.3-eksbuild.4\\\":
ErrImagePull: failed to pull and unpack image \\\"602401143452.dkr.ecr.us-
west-2.amazonaws.com/eks/kube-proxy:v1.33.3-eksbuild.4\\\": failed to resolve reference \
\\"602401143452.dkr.ecr.us-west-2.amazonaws.com/eks/kube-proxy:v1.33.3-eksbuild.4\\\":
pull access denied, repository does not exist or may require authorization:
authorization failed: no basic auth credentials\"" pod="kube-system/kube-proxy-n98hn"
podUID="ce4cea0b-a2c1-4ff4-a6ed-0574d7068cbe"
```

To resolve this issue, add the EKS cluster node group role to the ECR.DKR and ECR.API custom policy.

The encryption case is failing with the following errors because of a missing CIDR range in the agent security group:

```
ERROR CCITaskUtils - Connection timed out (Connection timed out)
2026-02-02 08:17:19Z ERROR CCITaskUtils - ***
2026-02-02 08:17:19Z ERROR CCITaskUtils - KMS Service connection failed with stacktrace
 : java.net.ConnectException: Connection timed out (Connection timed out)
```

To resolve this issue, add both the subnets CIDR range to the Secure Agent security group.

Spot instance pricing isn't working in my advanced cluster.

Spot pricing needs internet access to reach the AWS Pricing API. If your organization uses AWS PrivateLink to connect to IDMC, there is no access to the public internet and network access is restricted to the AWS private endpoints. This causes the cluster to disable spot instances and default to an on-demand procurement for all nodes.

Author

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